

It is allowed to have one electronic device to stream yourself via Zoom (this may be a smartphone in airplane mode) and one other device (but no smartphone) to display the exam. This exam consists of 6 questions on 3 pages (36 points).

Please write all answers in English. Grade = $\frac{\text{total}+4}{4}$.

You have 165 minutes to write the exam.

GOOD LUCK!

Question 1 [6 points]

Indicate for each statement whether it is correct or not. Motivate/explain your answers shortly; also if you choose “correct”, show with your explanation that you know why it is correct.

Note: just “correct” or “false” without further explanation will result in at most 0.5 point for each question.

- [2 points] For chi-square goodness-of-fit tests, the degrees of freedom depend on the sample size.
- [2 points] The rectangular kernel is generally a good choice for kernel density estimation.
- [2 points] t -tests can be used to detect outliers in the context of linear regression models.

Question 2 [5.5 points]

Let $X_1, X_2, \dots, X_n$ ($n \geq 2$) be independent random variables with cumulative distribution function F .

- [1 point] Explain in your own words what a nonparametric test is.
- [1.5 points] State the null and alternative hypotheses that can be tested with the Shapiro-Wilk test.
- [3 points] Describe the steps that are made in a parametric bootstrap procedure for the estimation of the distribution of a statistic $T(X_1, \dots, X_n)$ if the distribution of $X_1, \dots, X_n$ depends on some parameter θ , i.e. $F = F_\theta$.

Question 3 [6 points]

In this question we let the significance level for all hypothesis tests be $\alpha = 5\%$. Let $X_1, X_2, \dots, X_n$ ($n \geq 2$) be independent random variables that follow an unknown symmetric distribution P with mean $\theta \in \mathbb{R}$ and variance $\sigma^2 > 0$. A scale-invariant test for $H_0 : \theta = 0$ against $H_1 : \theta \neq 0$ can be based on the test statistic

$$T(X_1, \dots, X_n) = \frac{|\text{med}(X_1, \dots, X_n) - 0|}{\text{mad}(X_1, \dots, X_n)},$$

where med and mad denote the median and the median absolute deviation statistics, respectively.

- [3 points] We choose the following bootstrap version of the test statistic:

$$T^* = \frac{|\text{med}(X_1^*, \dots, X_n^*) - \text{med}(X_1, \dots, X_n)|}{\text{mad}(X_1^*, \dots, X_n^*)}.$$

Describe the steps for obtaining suitable bootstrap samples $X_1^*, \dots, X_n^*$, for conducting the bootstrap test, and for drawing a conclusion.

- [1.5 points] Computer simulations indicated that the asymptotic relative efficiency of the two-sample t -test with respect to the bootstrap test from a. is about 1.1 if the data come from a t -distribution with 10 degrees of freedom shifted by $\theta = 0.2$. Explain the meaning of this asymptotic relative efficiency in terms of sample sizes.

Now consider a one-sided test for $H_0 : \theta \leq 0$ against $H_1 : \theta > 0$ and another one for $H_0 : \theta \geq 0$ against $H_1 : \theta < 0$, and their one-sided p -values p_ℓ and p_r , respectively.

- [1.5 point] Explain how a p -value for a two-sided test for $H_0 : \theta = 0$ against $H_1 : \theta \neq 0$ can be obtained from the one-sided p -values p_ℓ and p_r .

Question 4 [6 points] Let $X_1, \dots, X_n$ be independent and identically distributed random variables with density function f and cumulative distribution function F .

- [1 point] Explain why the Epanechnikov kernel is generally a good choice for kernel density estimation.
- [1 point] Consider the following two approaches for the estimation of $F(t)$, $t \in \mathbb{R}$:
 - integrate the kernel density estimator based on the Gaussian kernel from $-\infty$ to t ;
 - use the empirical distribution function, i.e. $\hat{F}_n(t)$.
 Explain one advantage of Approach 1. over Approach 2. or the other way around.

Suppose now that the dataset $\mathbf{x}$ contains realizations of the random variables $X_1, \dots, X_n$. Figure 1 below displays a histogram, symplot, and various QQ-plots of $\mathbf{x}$.

- [2 points] Based on the plots in Figure 1, which location-scale family seems most appropriate for these data? Motivate your answer based on the histogram, the symplot, and the QQ-plots.
Note: you do not need to specify the location and scale parameters.
- [2 points] Consider the bootstrap modified Kolmogorov-Smirnov test¹ at level $\alpha=5\%$ for testing the null hypothesis that the real distribution of the data belongs to the location-scale family of the χ^2 -distribution function with four degrees of freedom. The p -value is $p=0.1124$. Interpret the implications of this outcome, also what this means for the use of χ^2_4 as a possible model distribution for the data.

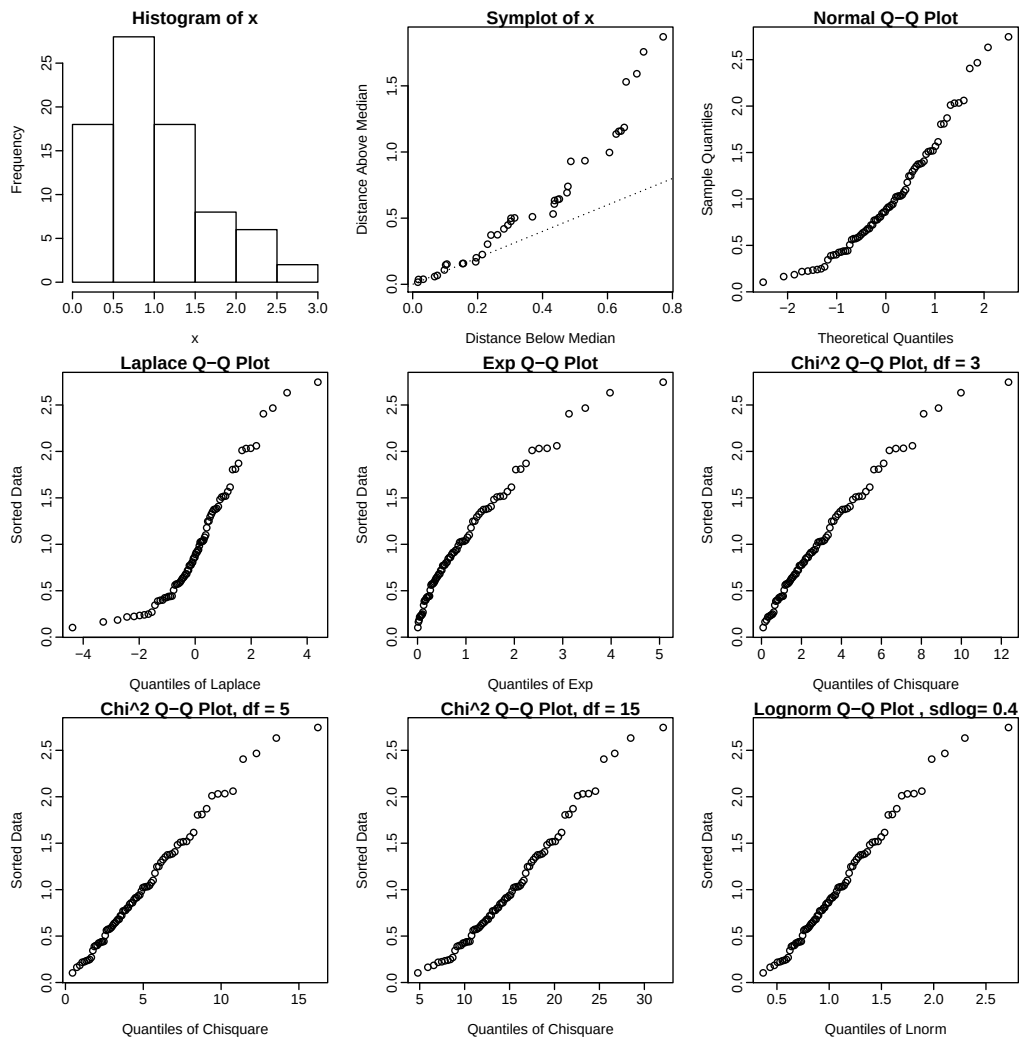


Figure 1: Histogram, symplot, and QQ-plots against indicated distributions of a sample $\mathbf{x}$ of size $n = 80$.

¹In the test statistic, the reference distribution function F_0 is shifted and rescaled by means of the sample mean and sample standard deviation, respectively, of the sample $\mathbf{x}$.

Question 5 [6.5 points]

In a study about the effects of different diets, rats were randomized into two different groups and given two different diets. After a while the liver weight (as % of body weight) was measured and then categorized ($\leq 3.7\%$ or $> 3.7\%$). The data is summarized in the following table:

liver weight / diet	1	2	total
$\leq 3.7\%$	6	3	9
$> 3.7\%$	2	4	6
total	8	7	15

- [3 points] How many samples are involved in this study? Motivate your answer. Also, state a set of null and alternative hypotheses that can be tested in the present context.
- [1.5 points] There is an exact test for the hypotheses from a. State the name of that test, the value of its test statistic for the dataset at hand, and the distribution of the test statistic under the null hypothesis.
You do not need to specify the exact parameters of the distribution under the null hypothesis.
- [2 points] Sketch all steps that are necessary to identify extreme cell entries in a general contingency table if the sample sizes are large enough.
Note: you do not need to write down explicit formulas nor actually find the extreme entries.

Question 6 [6 points]

- [3 points] Explain in general how the step up variable selection method works.
- [2 points] Suppose we wish to set up a linear regression model for a certain response variable Y ; possible explanatory variables are X_1 and X_2 . Figure 2 below shows an added variable plot for X_2 in addition to X_1 (and the intercept) for a sample of size 50. Explain what precisely is plotted in Figure 2 and interpret the implications of this plot.
- [1 point] Explain what it means for a fitted linear regression model if the coefficient of determination is $R^2 = 0.86$.

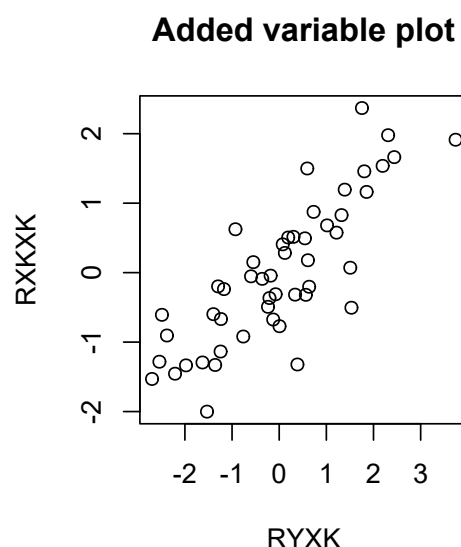


Figure 2: Added variable plot for the variable X_2 in addition to X_1 .